

Permutation With Repetition

$$n = \frac{n!}{s_1! s_2! \dots s_m!}$$

- 1) How many 9-digit numbers we can generate from digits 1, 1, 2, 2, 2, 3, 3, 3, 3?

$n = 9$ using above formula:

$$s_1 = 2$$

$$s_2 = 3$$

$$s_3 = 4$$

$$N = \frac{9!}{2! 3! 4!} = \frac{362880}{288}$$

$$N = 1260$$

We can generate 1260 different 9 digit numbers.

- 2) How many 9-digit numbers we can generate from digits 1, 1, 5, 5, 5, 0, 0, 0, 0?

$$n = 8$$

$$s_1 = 2$$

$$s_2 = 3$$

$$s_3 = 3$$

$$\text{Invalid numbers (IN)} = \frac{8!}{2! 3! 3!} = \frac{362880}{72} = 5040$$

$$IN = 5040$$

$$N_1 = N - IN$$

$$1260 - 5040 = 700$$

$$N_1 = 700$$

We can generate 700 different 9 digit numbers from 1, 1, 5, 5, 5, 0, 0, 0, 0.

Variations

$$V_r(n) = \frac{n!}{(n-r)!} = \binom{n}{r} \cdot r!$$

1.) How many valid 3-digit numbers, where each digit is unique, can be generated from digits 1, 2, 3, 4, 5, 6, 7, 8, 9?

$$n = 9$$

$$r = 3$$

$$N = \frac{n!}{(n-r)!} = \frac{9!}{(9-3)!} = \frac{9!}{6!} = \frac{362880}{720}$$

$$N = 504$$

We can generate 504 valid 3-digit numbers from digits 1, 2, 3, 4, 5, 6, 7, 8, 9.

2.) How many valid 3-digit numbers, where each digit is unique, can be generated from digits 0, 1, 2, 3, 4, 5, 6, 7, 8?

$n_1 = 8$ (1st digit cannot be zero, so we have 8 digit)

$$n_2 = 8$$

$$n_3 = 7$$

$$N = n_1 \times n_2 \times n_3 = 8 \times 8 \times 7$$

$$N = 448$$

We can generate 448 diff 3 digit numbers from 0, 1, 2, 3, 4, 5, 6, 7, 8.

Variations with Repetition

(2)

$$V_r(n) = n^r$$

- 1) How many valid 3-digit numbers can be generated from digits 1, 2, 3, 4, 5, 6, 7, 8, 9?

$$n = 9.$$

$$r = 3$$

$$N = 9^3 = 9 \times 9 \times 9.$$

$$\boxed{N = 729}$$

- 2) How many valid 3-digit numbers can be generated from digits 0, 1, 2, 3, 4, 5, 6, 7, 8?

$$n_1 = 8$$

$$n_2 = 9$$

$$n_3 = 9.$$

(1st digit cannot be zero)
(can be zero)
(-11 -)

$$N = n_1 \times n_2 \times n_3 = 8 \times 9 \times 9.$$

$$\boxed{N = 648}$$

we can generate 648 d/f 3 digit numbers from digits 0, 1, 2, 3, 4, 5, 6, 7, 8.

Combinations

$$C_r(n) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- i) What is your chance to win 1st prize for 20 Euro?
x How ~~add~~ much money you have to spend to have 10% chance to win 1st prize.

$n = 10$ (lottery bag 0 to 9)

$r = 4$ (ticket = 4 numbers)

Total no. of combination is.

$$N = \frac{n!}{r!(n-r)!} = \frac{10!}{4!(10-4)!} = \frac{10!}{4!6!} = \frac{3628800}{17280}$$

$$N = 210$$

Total number of tickets

Probability of winning first prize is $1/N$.

$$= 1/210 = 0.00476190476190$$

Probability of winning 1st prize for 20 euro is.

$$(20/N) * 100 = (20/210) * 100 = 9.52380952380952\%$$

Probability

Money spent to have 10% chance to win the 1st prize

$$N * (10/100) = 210 * (10/100) = 21 \text{ Euro}$$

Combinations with Repetition

$$\bar{C}_r(n) = \binom{n+r-1}{r}$$

$n = 10$

$r = 4$

Total no. of combinations is.

$$N = \frac{(n+r-1)!}{r!(n-1)!} = \frac{(10+4-1)!}{4!(10-1)!} = \frac{13!}{4!9!}$$

$$N = 715$$

Total number of tickets

Probability of winning first prize. $1/N$

$$1/75 = \boxed{0.00139860139860}$$

Probability of winning 1st prize for 20 euro is

$$\left(\frac{20}{N}\right) * 100 = \boxed{2.79720279720279\%}$$

Money spent to have 100% chance to win the 1st prize.

$$N * (10/100) = 75 (10/100) = \boxed{7.5 \text{ euros}}$$

Example L3

$$p = 0.2 \quad p \in \{0,1\}, p < 1$$

$$N = 20$$

$x \in \{0, N\} \in \mathbb{W}$, i.e., $x = 0, 1, 2, 3, \dots, N$ (failed computers)

2) what is the probability that no computer will need technical support after update?

$$(1-p)^N = (1-0.2)^{20} = \boxed{0.01152921504606}$$

3) what is the probability that all computers will need technical support after update?

$$p^N = (0.2)^{20} = \boxed{1.048576 \text{E}-14}$$

4) what is the probability that only one particular computer will need technical support after update.

$$p * (1-p)^{(N-1)} = (0.2) * (1-0.2)^{19} = \boxed{0.00288230376151}$$

5) What is the probability that one computer (anyone out of all) will need technical support after update?

$$Np(1-p)^{(N-1)} = 20 \times (0.2)(1-0.2)^{(20-1)}$$

$$= \boxed{0.05764607523034}$$

6) What is the probability that any two computers will need technical support after update?

$$\frac{N(N-1)}{2} p^2(1-p)^{(N-2)}$$

$\frac{N(N-1)}{2}$ = No. of ways to choose 2 items from N items

$$\frac{20(20-1)}{2} (0.2)^2 (1-0.2)^{(20-2)}$$

$$190(0.04)(1-0.2)^{18} = \boxed{0.13690942867206}$$

7) What is the probability that any three computers will need technical support after update?

$$\frac{N(N-1)(N-2)}{6} p^3(1-p)^{(N-3)}$$

$$\frac{20(20-1)(20-2)}{6} (0.2)^3 (1-0.2)^{(20-3)}$$

$$= 1140(0.008)(1-0.2)^{17} = \boxed{0.20536414300809}$$

8) What is variance & S.D of this probability distribution? how many computers is estimated to need support plus minus? mean $\pm$ Variance.

$$\text{Mean} = N \times p = 20 \times 0.2 = \boxed{\text{Mean} = 4}$$

$$\text{Variance} = N \times p \times (1-p) = 20 \times 0.2 \times (1-0.2) = \boxed{3.2}$$

$$\text{Standard deviation (S.D)} = \sqrt{N \times p \times (1-p)} = \sqrt{3.2} = \boxed{1.78885438}$$

$$\text{computers estimated} : N \times p \pm \sqrt{N \times p \times (1-p)} = 4 \pm \sqrt{3.2}$$

$$= \boxed{4 \pm 1.78885438}$$